

Section I

NESA Number:

10 marks

--	--	--	--	--	--	--	--

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

1. What is the domain and range of $y = 2 \sin^{-1} \frac{2x}{5}$?

(A) Domain: $-\frac{5}{2} \leq x \leq \frac{5}{2}$, Range: $-\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$.

(B) Domain: $-\frac{2}{5} \leq x \leq \frac{2}{5}$, Range: $-\pi \leq y \leq \pi$.

(C) Domain: $-\frac{5}{2} \leq x \leq \frac{5}{2}$, Range: $-\pi \leq y \leq \pi$.

(D) Domain: $-\frac{2}{5} \leq x \leq \frac{2}{5}$, Range: $-\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$

2. Given $\overrightarrow{OA} = -2\hat{i} + 3\hat{j}$ and $\overrightarrow{AB} = 4\hat{i} - \hat{j}$, which is the correct value for $\overrightarrow{OB}$?

(A) $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$ (B) $\begin{pmatrix} -6 \\ 2 \end{pmatrix}$

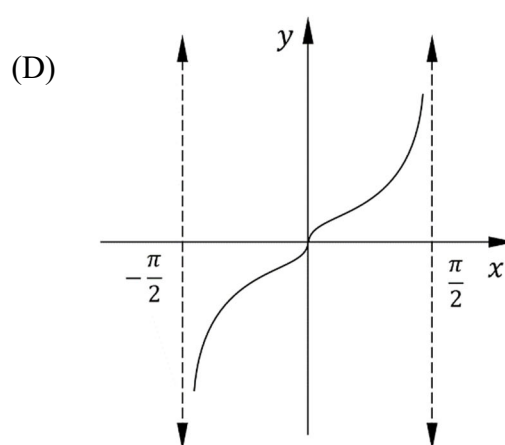
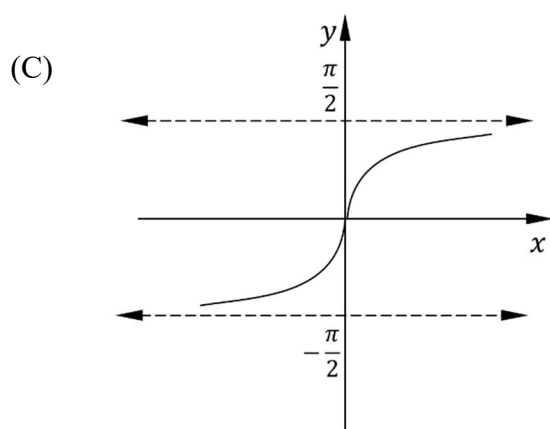
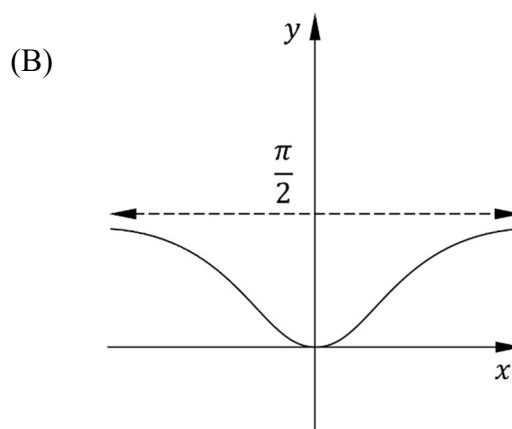
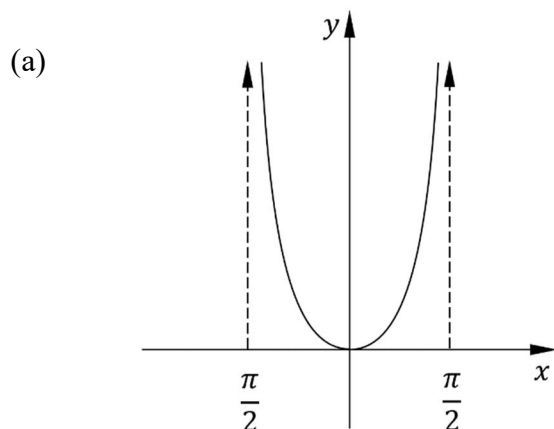
(C) $\begin{pmatrix} 6 \\ -4 \end{pmatrix}$ (D) $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$

3. What is the remainder when $P(x) = x^3 - 2x + 3$ is divided by $(2x - 1)$

(A) $3\frac{7}{8}$ (B) $2\frac{1}{8}$

(C) 2 (D) 4

4. Which of the following graphs best shows $y = \tan^{-1}(x^2)$?



5. Consider the differential equation $\frac{dy}{dx} = 4xy$.

Which of the following is the family of solutions to the equation.

(A) $y = Ae^{2x^2}$

(B) $y = \ln(2x^2) + c$

(C) $y = 2x^2 \ln|y| + c$

(D) $y = 4x \ln|y| + c$

6. The cartesian equation of the curve with the parametric equations $x = 2e^t$ and $y = \cos(1 + e^{3t})$ for $0 \leq t \leq \frac{3}{4}$ is given by:

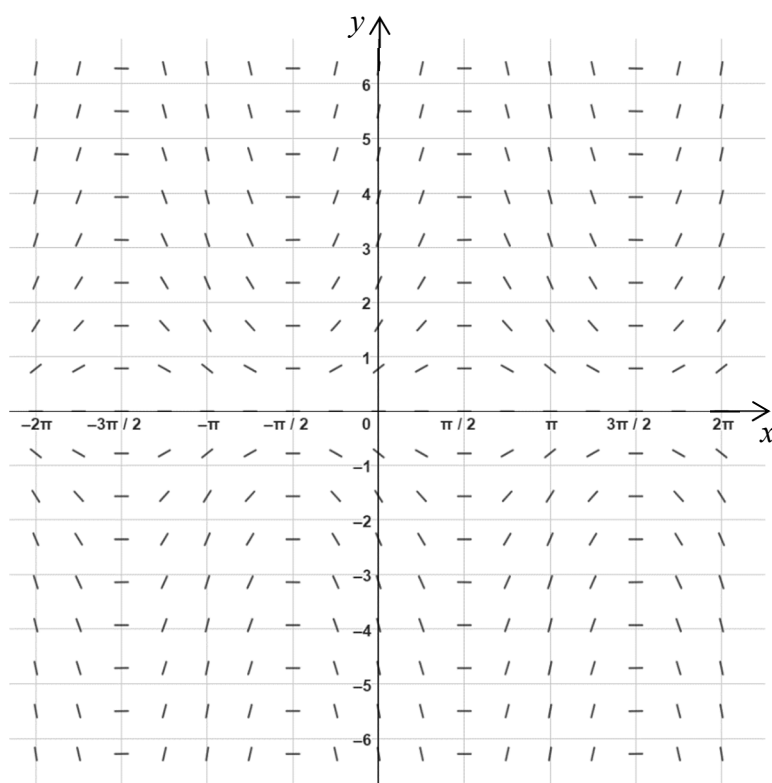
(A) $y = \cos\left(1 + \frac{e^3}{8}x\right)$

(B) $y = \cos\left(1 + \frac{x}{2}\right)$

(C) $y = \cos\left(1 + \frac{x}{2} + e^3\right)$

(D) $y = \cos\left(1 + \frac{x^3}{8}\right)$

7. Which differential equation is shown in the slopefield below?



(A) $y' = y \cos x$

(B) $y' = y \sin x$

(C) $y' = x \cos y$

(D) $y' = x \sin y$

8. What is the value of k such that $\int_0^k \frac{1}{\sqrt{4-9x^2}} dx = \frac{\pi}{18}$

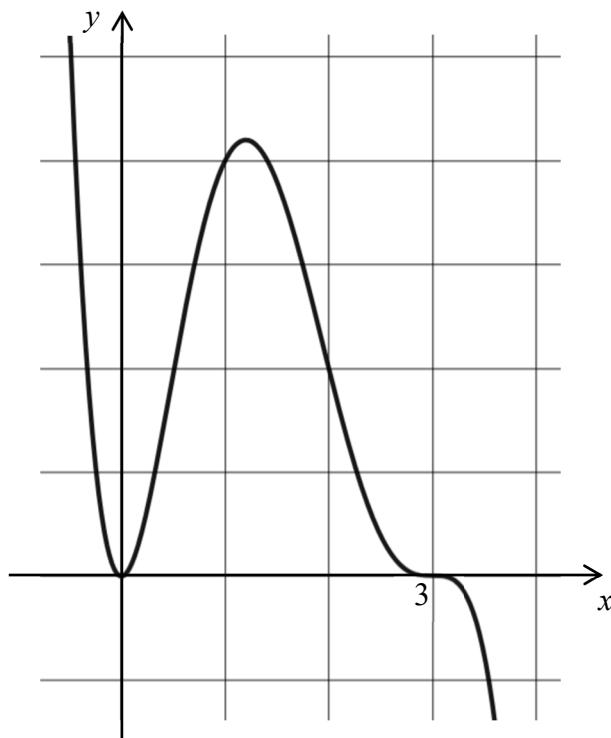
(A) -3

(B) $\frac{1}{3}$

(C) $-\frac{1}{3}$

(D) 3

9. Which of the following could be the polynomial $y = P(x)$.



(A) $y = x^3(x-3)^2$

(B) $y = x^2(x-3)^3$

(C) $y = -x^3(x-3)^2$

(D) $y = -x^2(x-3)^3$

10. The integral $\int_0^{\frac{\pi}{8}} \cos 6x \cos 2x \, dx$ simplified is equal to:

(A) $\frac{3}{16}$

(B) $\frac{1}{8}$

(C) 0

(D) $\frac{1}{16}$

Section II

60 marks

Attempt Questions 1 – 4

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available. In questions 11 – 14, your responses should include relevant mathematical reasoning and/or calculations.

Question 1 (16 marks) - Start your work in Question 1 Answer Booklet

(a) If $\underline{a} = 3\underline{i} - 2\underline{j}$ and $\underline{b} = -\underline{i} + 4\underline{j}$, calculate:

(i) $\underline{b} - \underline{a}$ 1

(ii) $\underline{a} \cdot \underline{b}$ 1

(b) Differentiate $y = \frac{1}{3} \tan^{-1} 3x$. 2

(c) Find $\int \frac{1}{x^2 + 2x + 5} dx$ 2

(d) Evaluate $\int_0^{\frac{1}{2}} \sqrt{\frac{x}{1-x}} dx$ using the substitution $x = \sin^2 \theta$. 3

(e) (i) If the polynomials $P(x) = 2x^3 + mx^2 + 2x - 3$ and $Q(x) = x^2 + nx - 3$ have the same remainder when divided by $x + 2$, write an expression for m in terms of n . 2

(ii) Given that $(x - 3)$ is a factor of $Q(x)$, find the value of m and n . 2

(f) Find the exact value of $\cos \frac{\pi}{8}$ giving your answer in simplest form. 3

End of Question 1.

Question 2 (16 marks) - Start your work in Question 2 Answer Booklet

- (a) Prove, by Mathematical Induction, that $5^{2n+1} + 2^{2n+1}$ is divisible by 7 for all integers $n \geq 1$. **3**
- (b) In a 16 member soccer squad 12 are right-handed while 4 are left-handed. If 11 members are to be selected as the starting line up (players participating at the start of a game), in how many ways will there be at least three left-handed players in the starting line up? **2**
- (c) The functions f and g are defined by $f(x) = \sqrt{4-x^2}$ and $g(x) = x-1$.
- (i) Show that the domain of $f(g(x))$ is $-1 \leq x \leq 3$. **1**
- (ii) Hence state the range of the function $f(g(x))$. **1**
- (iii) What is the largest domain which includes the point $(3, 0)$ over which $f(g(x))$ has an inverse function? **1**
- (iv) Hence find $h(x)$, the inverse function of the composite function $f(g(x))$, stating its domain and range. **3**
- (v) Sketch the graph of $y = h(x)$. **2**
- (d) Show that $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi$ **3**

End of Question 2.

Question 3 (15 marks) - Start your work in Question 3 Answer Booklet

(a) For vectors $\underline{u} = 3\underline{i} + b\underline{j}$ and $\underline{v} = -\underline{i} - 3\underline{j}$

(i) Write an expression for the projection of vector u onto vector v . 2

(ii) Given that the length of this projection is 3 units, find the value of b . 2

(b) Find in the form $y = f(x)$ the solution of the differential equation $y' = \frac{2e^{-\frac{1}{2}y}}{\cos^2 x}$,
given that $y = \ln 3$ when $x = \frac{\pi}{3}$. 3

(c) Newton's law of cooling states that when an object at temperature $T^\circ\text{C}$ is placed in an environment at temperature $A^\circ\text{C}$, the rate of temperature loss is given by the equation:

$$\frac{dT}{dt} = -k(T - A) \quad \text{where } t \text{ is the time in minutes and } k \text{ is a positive constant.}$$

(i) Use differential equations to show that $T = A + Be^{-kt}$ is a solution to the above equation. 1

(ii) A cup of tea with initial temperature of 90°C is placed in a room in which the surrounding temperature is maintained at 25°C . After 25 minutes, the temperature of the cup of tea is 45°C . How long will it take for the it's temperature to reduce to 30°C ? Answer correct to the nearest minute. 3

(d) (i) Ten friends are going to be divided into groups of 5, 3 and 2 members as part of a competition. In how many ways can the groups be formed? 2

(ii) If n friends are divided into groups of made up of c , k , and r members
where $c + r + k = n$ and $c > k > r$. Explain why

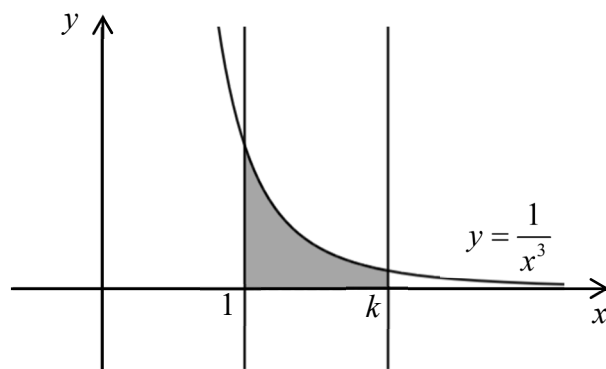
$$\binom{n}{n-k-r} \binom{k+r}{r} = \binom{n}{r} \binom{n-r}{k}. \quad \text{2}$$

End of Question 3.

Question 4 (13 marks) - Start your work in Question 4 Answer Booklet

- (a) A spherical ball is expanding so that its volume is increasing at the constant rate of 10 mm^3 per seconds. What is the rate of increase of the radius when the surface area is 400 mm^2 ? **2**

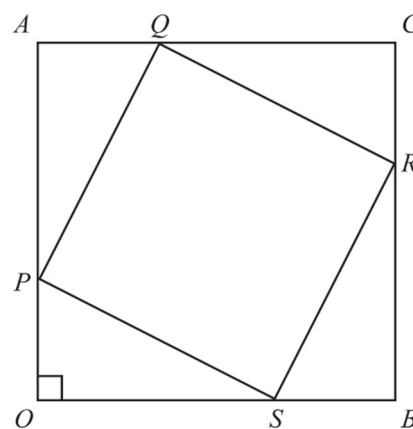
- (b) The graph of $y = \frac{1}{x^3} \{x > 0\}$ is shown below. The shaded area is rotated about the y-axis.



- (i) Show that the generated volume in terms of k is $V = \left(2\pi - \frac{2\pi}{k}\right) \text{ units}^3$. **4**
- (ii) Explain what happens to the volume as $k \rightarrow \infty$. **1**
- (iii) If the volume of the solid form is $\frac{3\pi}{2} \text{ units}^3$, find the value of k . **1**

- (c) Consider the square $OACB$ where point O is the origin. Let the position vector of points A and B be defined as $\underline{a}$ and $\underline{b}$ respectively i.e. $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$.

Let points P, Q, R and S be defined so that $\overrightarrow{OP} = k\underline{a}$, $\overrightarrow{AQ} = k\underline{b}$, $\overrightarrow{RC} = k\underline{a}$ and $\overrightarrow{SB} = k\underline{b}$ where $0 \leq k \leq 1$. This means points P, Q, R and S are positioned along their respective sides in equal proportions.



Use vector methods to prove that the size of $\angle PQR = 90^\circ$. **5**

End of Examination.

Section I - Solutions

NESA Number:

--	--	--	--	--	--	--	--

10 marks

Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

1. What is the domain and range of $y = 2\sin^{-1}\frac{2x}{5}$?

(A) Domain: $-\frac{5}{2} \leq x \leq \frac{5}{2}$, Range: $-\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$.

(B) Domain: $-\frac{2}{5} \leq x \leq \frac{2}{5}$, Range: $-\pi \leq y \leq \pi$.

☒ (C) Domain: $-\frac{5}{2} \leq x \leq \frac{5}{2}$, Range: $-\pi \leq y \leq \pi$.

(D) Domain: $-\frac{2}{5} \leq x \leq \frac{2}{5}$, Range: $-\frac{\pi}{4} \leq y \leq \frac{\pi}{4}$

2. Given $\overrightarrow{OA} = -2\mathbf{i} + 3\mathbf{j}$ and $\overrightarrow{AB} = 4\mathbf{i} - \mathbf{j}$, which is the correct value for $\overrightarrow{OB}$?

(A) $\begin{pmatrix} 2 \\ -4 \end{pmatrix}$ (B) $\begin{pmatrix} -6 \\ 2 \end{pmatrix}$

(C) $\begin{pmatrix} 6 \\ -4 \end{pmatrix}$ ☒ (D) $\begin{pmatrix} 2 \\ 2 \end{pmatrix}$

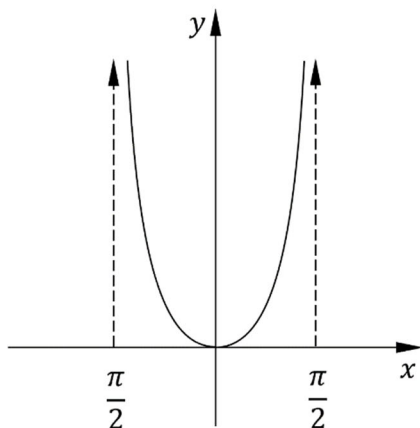
3. What is the remainder when $P(x) = x^3 - 2x + 3$ is divided by $(2x - 1)$

(A) $3\frac{7}{8}$ ☒ (B) $2\frac{1}{8}$

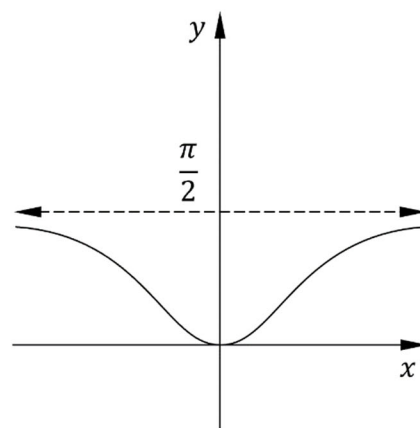
(C) 2 (D) 4

4. Which of the following graphs best shows $y = (x^2)$?

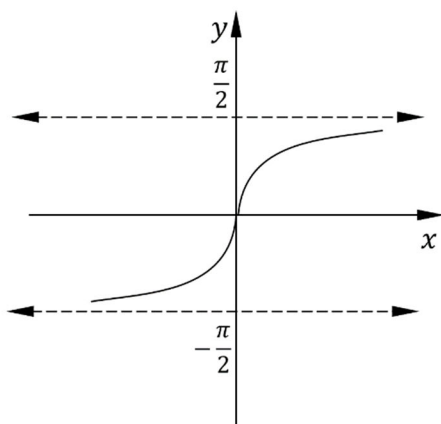
(a)



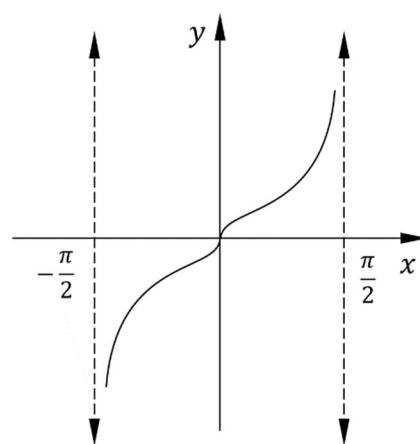
(B)



(C)



(D)



5. Consider the differential equation $\frac{dy}{dx} = 4xy$.

Which of the following is the family of solutions to the equation.

(A)

$$y = Ae^{2x^2}$$

(B)

$$y = \ln(2x^2) + c$$

(C)

$$y = 2x^2 \ln|y| + c$$

(D)

$$y = 4x \ln|y| + c$$

6. The cartesian equation of the curve with the parametric equations $x = 2e^t$ and $y = \cos(1 + e^{3t})$ for $0 \leq t \leq \frac{3}{4}$ is given by:

(A)

$$y = \cos\left(1 + \frac{e^3}{8}x\right)$$

(B)

$$y = \cos\left(1 + \frac{x}{2}\right)$$

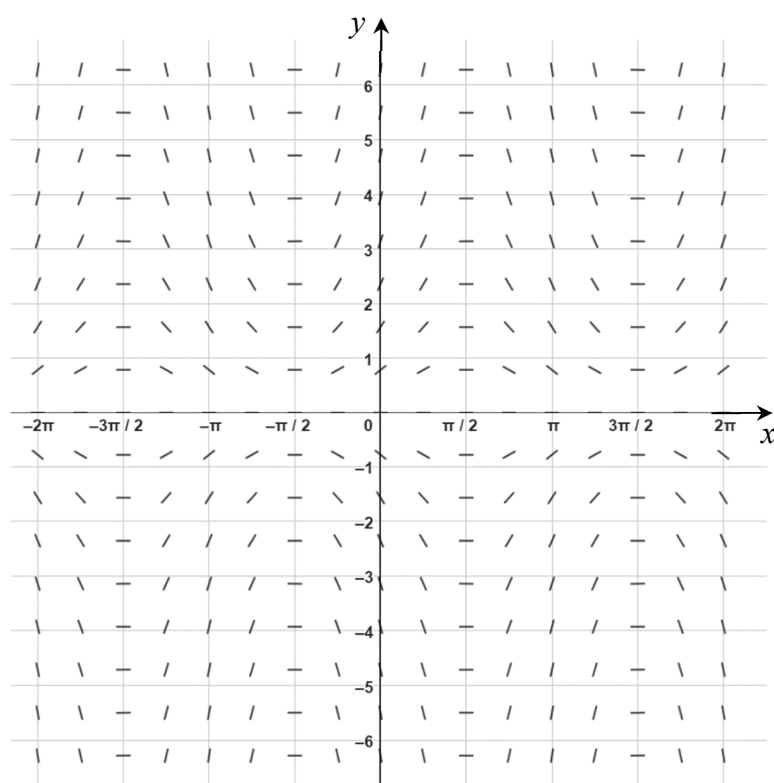
(C)

$$y = \cos\left(1 + \frac{x}{2} + e^3\right)$$

(D)

$$y = \cos\left(1 + \frac{x^3}{8}\right)$$

7. Which differential equation is shown in the slopefield below?



(A) $y' = y \cos x$

(B) $y' = y \sin x$

(C) $y' = x \cos y$

(D) $y' = x \sin y$

8. What is the value of k such that $\int_0^k \frac{1}{\sqrt{4-9x^2}} dx = \frac{\pi}{18}$

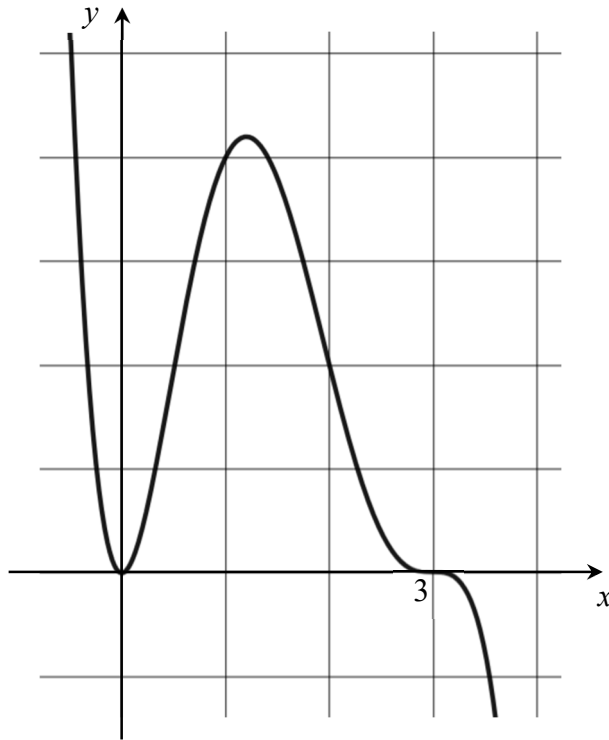
(A) -3

(B) $\frac{1}{3}$

(C) $-\frac{1}{3}$

(D) 3

9. Which of the following could be the polynomial $y = P(x)$.



(A) $y = x^3(x-3)^2$

(B) $y = x^2(x-3)^3$

(C) $y = -x^3(x-3)^2$

(D) $y = -x^2(x - 3)^3$

10. The integral $\int_0^{\frac{\pi}{8}} \cos 6x \cos 2x dx$ simplified is equal to:

(A) $\frac{3}{16}$

(B) $\frac{1}{8}$

(C) 0

(D) $\frac{1}{16}$

Section II

Question 1 (16marks) - Start your work in Question 1 Answer Booklet

(a) If $\underline{a} = 3\underline{i} - 2\underline{j}$ and $\underline{b} = -\underline{i} + 4\underline{j}$, calculate:

(i) $\underline{b} - \underline{a}$

1

$\underline{b} - \underline{a} = \begin{bmatrix} -1 \\ 4 \end{bmatrix} - \begin{bmatrix} 3 \\ -2 \end{bmatrix}$ $= -4\underline{i} + 6\underline{j}$	1 – for answer Marker's Comments: Answered well.
--	--

(ii) $\underline{a} \cdot \underline{b}$

1

$\underline{a} \cdot \underline{b} = (3)(-1) + (-2)(4)$ $= -3 - 8$ $= -11$	1 – for answer Marker's Comments: Answered well.
--	--

(b) Differentiate $y = \frac{1}{3} \tan^{-1} 3x$.

2

$y = \frac{1}{3} \tan^{-1} 3x$ $y' = \frac{1}{3} \frac{3}{1+9x^2}$ $= \frac{1}{1+9x^2}$	1 – for differentiation 1 – for simplified answer Marker's Comments: Answered well.
---	--

(c) Find $\int \frac{1}{x^2 + 2x + 5} dx$

2

$\int \frac{1}{x^2 + 2x + 5} dx = \int \frac{1}{(x+1)^2 + 2^2} dx$ $= \frac{1}{2} \tan^{-1} \frac{x+1}{2} + C$	1 – for rearrangement 1 – answer Marker's Comments: Most students who new the strategy got full marks. Those who did not used either logs or attempted to incorrectly split the fraction, both leading to incorrect answers.
--	---

(d) Evaluate $\int_0^{\frac{1}{2}} \sqrt{\frac{x}{1-x}} dx$ using the substitution $x = \sin^2 \theta$.

3

$x = \sin^2 \theta$ $\frac{dx}{d\theta} = 2 \sin \theta \cos \theta$ when $x = 0$ $\theta = 0$ $x = \frac{1}{2}$ $\theta = \frac{\pi}{4}$ $\therefore \int_0^{\frac{1}{2}} \sqrt{\frac{x}{1-x}} dx = \int_0^{\frac{\pi}{4}} \sqrt{\frac{\sin^2 \theta}{1 - \sin^2 \theta}} 2 \sin \theta \cos \theta d\theta$ $= \int_0^{\frac{\pi}{4}} \sqrt{\frac{\sin^2 \theta}{\cos^2 \theta}} 2 \sin \theta \cos \theta d\theta$ $= \int_0^{\frac{\pi}{4}} 2 \sin^2 \theta d\theta \quad [\cos 2\theta = 1 - 2\sin^2 \theta]$ $= \int_0^{\frac{\pi}{4}} (1 - \cos 2\theta) d\theta$ $= \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{4}}$ $= \left(\frac{\pi}{4} - \frac{1}{2} \sin \frac{\pi}{2} \right) - \left(0 - \frac{1}{2} \sin 0 \right)$ $= \frac{\pi}{4} - \frac{1}{2}$	1 – rewriting the integral in terms of theta 1 – integrating 1 – final exact answer. Marker's comments: Many students struggled with this question. Most students made the question more complicated by not using the basic chain rule when differentiating $x = \sin^2 \theta$.
---	---

(e) (i) If the polynomials $P(x) = 2x^3 + mx^2 + 2x - 3$ and $Q(x) = x^2 + nx - 3$ have the same remainder when divided by $x + 2$, write an expression for m in terms of n .

2

$P(-2) = -16 + 4m - 4 - 3$ $= 4m - 23$ $Q(-2) = 4 - 2n - 3$ $= 1 - 2n$ Now $P(-2) = Q(-2)$ $\Rightarrow 4m - 23 = 1 - 2n$ $4m = 24 - 2n$ $m = 6 - \frac{n}{2}$	1 – for equating the remainders 1 – for the expression Marker's Comments: Answered well. Students who used the remainder theorem were mostly successful. Those who used long division to find the remainders, made mistakes, leading to incorrect answers.
--	--

(ii) Given that $(x-3)$ is a factor of $Q(x)$, find the value of m and n .

2

$Q(3) = 9 + 3n - 3$ $\therefore 3n + 6 = 0$ $n = -2$ $m = 8$		1 – for value of n 1 – value of m Marker's Comments: Answered well.
--	--	---

(f) Find the exact value of $\cos \frac{\pi}{8}$ giving your answer in simplest form.

3

$\cos 2\theta = 2\cos^2 \theta - 1$ $\Rightarrow \cos^2 \theta = \frac{1}{2} [\cos 2\theta + 1]$ $\therefore \cos^2 \frac{\pi}{8} = \frac{1}{2} \left(\cos \frac{\pi}{4} + 1 \right)$ $= \frac{1}{2} \left(\frac{1}{\sqrt{2}} + 1 \right)$ $= \frac{1}{2} \left(\frac{\sqrt{2}}{2} + 1 \right)$ $= \frac{\sqrt{2} + 2}{4}$ $\therefore \cos \frac{\pi}{8} = \frac{(\sqrt{2} + 2)^{1/2}}{2}; \left(\cos \frac{\pi}{8} > 0 \right)$ (alt answer: $\cos \frac{\pi}{8} = \left(\frac{1}{\sqrt{2}} + 1 \right)^{1/2}$)		1 – for establishing the initial relationship 1 – for correct expression 1 – simplified answer with reason for ignoring the negative case. Marker's comments: Generally answered well. Most students made a start achieving one mark. But a number of students ignored the negative case when finding the square root, losing the chance to explain why the positive case is the correct answer.
--	--	--

End of Question 1.

Question 2 (16 marks) - Start your work in Question 2 Answer Booklet

(a) Prove, by Mathematical Induction, that $5^{2n+1} + 2^{2n+1}$ is divisible by 7 for all

integers $n \geq 1$.

3

step 1: Prove for $n=1$

Proof: $5^{2+1} + 2^{2+1} = 5^3 + 2^3$
 $= 133$
 $= 7 \times 19$
 $\therefore$ true for $n=1$

step 2: Assume true for $n=k$ where k is a positive integer
i.e. $5^{2k+1} + 2^{2k+1} = 7M$ where M is an integer

Prove true for $n=k+1$
i.e. Prove $5^{2k+3} + 2^{2k+3} = 7N$ where N is an integer

Proof:

$$\begin{aligned} \text{LHS} &= 5^2 \times 5^{2k+1} + 2^{2k+3} \\ &= 5^2 (7M - 2^{2k+1}) + 4 \times 2^{2k+1} \quad (\text{by assumption}) \\ &= 7 \times 25M + (4 - 25) 2^{2k+1} \\ &= 7 \times 25M - 7 \times 3(2^{2k+1}) \\ &= 7(25M - 3 \times 2^{2k+1}) \\ &= 7N \quad \text{where } N = 25M - 3 \times 2^{2k+1} \\ &= \text{RHS} \end{aligned}$$

step 3: Since true for $n=1$, by step 2 and mathematical induction, true for all positive integer n .

1 – proof for $n = 1$

1 – correct establishment of the assumption

1 – for the proof for $n = k+1$ case

Marker's comments:

Generally answered well but marks were deducted if k was not defined as a positive integer.

- (b) In a 16 member soccer squad 12 are right-handed while 4 are left-handed. If 11 members are to be selected as the starting line up (players participating at the start of a game), in how many ways will there be at least three left-handed players in the starting line up? **2**

$\binom{4}{3}\binom{12}{8} + \binom{4}{4}\binom{12}{7} = 1980 + 792 = 2772$	1 – for first case 1 – for final answer Marker's Comments: Generally answered well
---	--

- (c) The functions f and g are defined by $f(x) = \sqrt{4-x^2}$ and $g(x) = x-1$.

- (i) Show that the domain of $f(g(x))$ is $-1 \leq x \leq 3$. **1**

$f(g(x)) = \sqrt{4-(x-1)^2}$ $4 - (x-1)^2 \geq 0$ $(x-1)^2 \leq 4$ $-2 \leq x-1 \leq 2$ $D: \{-1 \leq x \leq 3\}$ Alternative approach: Domain of $f(x)$: $-2 \leq x \leq 2$ Domain of $f(g(x))$: $-2 \leq x-1 \leq 2$ $-1 \leq x \leq 3$	1 – steps towards finding the domain Marker's comments: Most students answered this question well, however, as this is a show question, students should include all necessary steps, including a stating that $4 - (x-1)^2 \geq 0$
---	---

- (ii) Hence state the range of the function $f(g(x))$. **1**

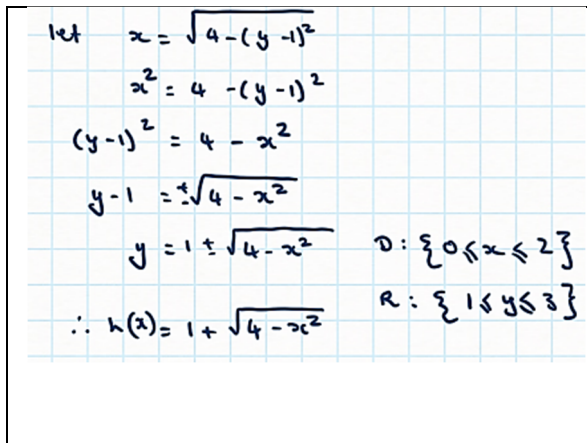
$R: \{0 \leq y \leq 2\}$	1 – range Marker's comments: Generally answered well.
--------------------------	--

- (iii) What is the largest domain which includes the point (3, 0) over which $f(g(x))$ has an inverse function? **1**

$D: \{1 \leq x \leq 3\}$	1 – domain Marker's comments: Generally answered well.
--------------------------	---

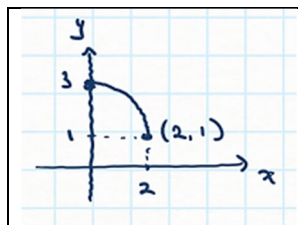
- (iv) Hence find $h(x)$, the inverse function of the composite function $f(g(x))$, stating its domain and range.

3

	1 – correct domain & range 1 – expression for y 1 – establishing the positive case as the function $h(x)$ Marker's comments: Students had difficulty identifying the correct domain and range. Many did not provide a rationale for taking the positive square root into their final answer. Had this been a show question a mark would have been deducted.
---	--

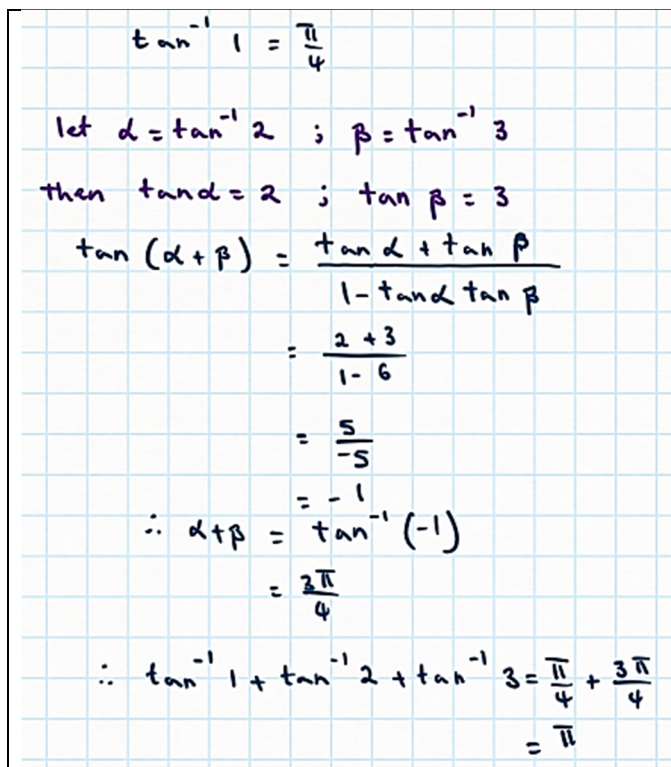
- (v) Sketch the graph of $y = h(x)$.

2

	1 – correct section of circle 1 – indicating the end points Marker's comments: Not answered very well due to the difficulties students had when answering part (iv)
--	--

- (d) Show that $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi$

3

	1 - $\tan^{-1} 1 = \frac{\pi}{4}$ 1 – establishing $\tan(\alpha + \beta)$ 1 – demonstrating $\text{LHS} = \pi$ Marker's comments: Not answered very well. Students did not readily recognise that $1 = \frac{\pi}{4}$ and as such had difficulty progressing with the solution. A few arrived at the correct proof but through very complicated approaches. Some students used the calculator to evaluate 2 and 3 which gave approximate values and consequently a mark was deducted. Others, incorrectly, tried applying the compound angle formula using three terms instead of two.
---	--

End of Question 2.

Question 3 (15 marks) - Start your work in Question 3 Answer Booklet

(a) For vectors $\underline{u} = 3\hat{i} + b\hat{j}$ and $\underline{v} = -\hat{i} - 3\hat{j}$

(i) Write an expression for the projection of vector $\underline{u}$ onto vector $\underline{v}$.

2

$\text{Proj}_{\underline{v}} \underline{u} = \frac{\underline{u} \cdot \underline{v}}{ \underline{v} ^2} \underline{v}$ $\underline{u} \cdot \underline{v} = (3)(-1) + (b)(-3)$ $= -3 - 3b$ $ \underline{v} ^2 = (-1)^2 + (-3)^2$ $= 10$ $\therefore \text{Proj}_{\underline{v}} \underline{u} = \frac{-3 - 3b}{10} (-\hat{i} - 3\hat{j})$ $= \frac{3(1+b)}{10} \hat{i} + \frac{9(1+b)}{10} \hat{j}$		1 – for $\underline{u} \cdot \underline{v}$ 1 – solution Marker's comment: Answered well.
--	--	---

(ii) Given that the length of this projection is 3 units, find the value of b .

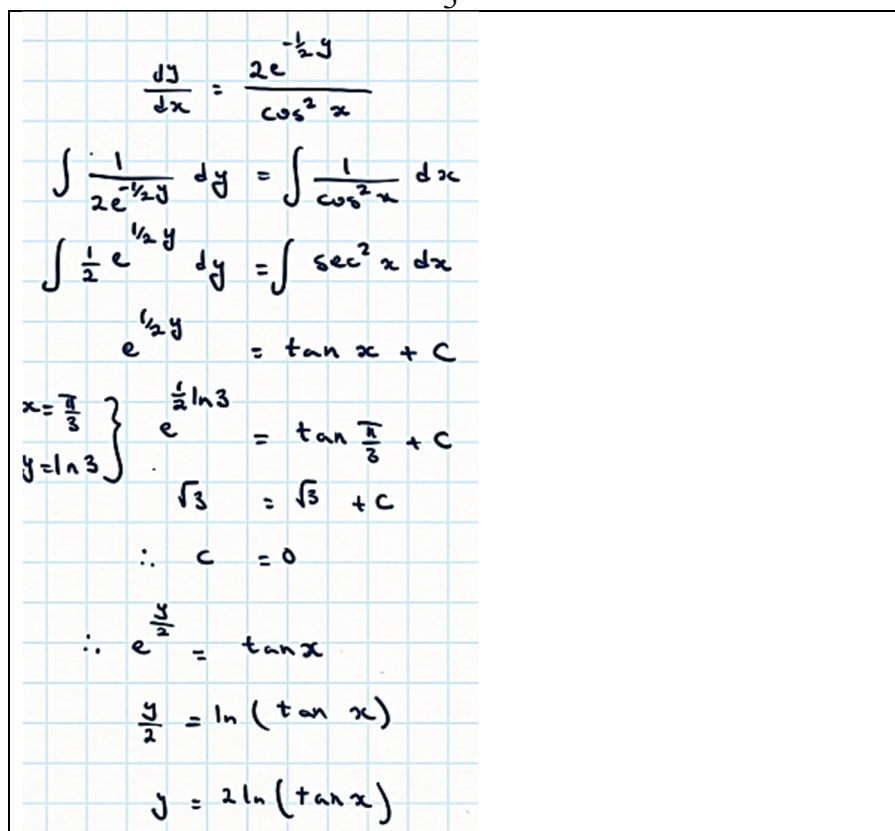
2

$ \text{Proj}_{\underline{v}} \underline{u} = \frac{\pm(1+b)}{10} \sqrt{9+81}$ $= \frac{\pm 3\sqrt{10}}{10} (1+b)$ $\therefore \frac{\pm 3\sqrt{10}}{10} (1+b) = 3$ $(1+b) = \pm \frac{10}{\sqrt{10}}$ $b = \pm \sqrt{10} - 1$	1 – length of projection in terms of b 1 – for answer Marker's comments: Many students forgot to do only the positive answer.
---	---

(b) Find in the form $y = f(x)$ the solution of the differential equation $y' = \frac{2e^{-\frac{1}{2}y}}{\cos^2 x}$,

given that $y = \ln 3$ when $x = \frac{\pi}{3}$.

3

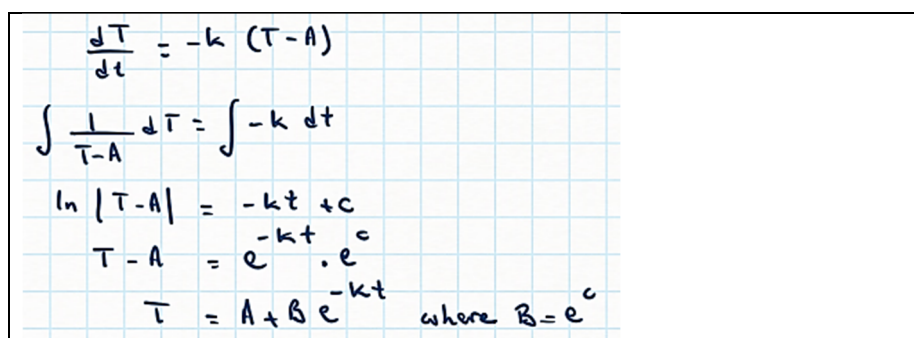
	1 – Establishing the integrals 1 – correctly integrating both integrals 1 – correct final expression for y Marker's comments: Setting of working was a problem, with many forgetting to include integral signs, or unable to provide the first integral statement at all. The integral of the exponential was done badly by many.
---	---

(c) Newton's law of cooling states that when an object at temperature $T^\circ \text{C}$ is placed in an environment at temperature $A^\circ \text{C}$, the rate of temperature loss is given by the equation:

$$\frac{dT}{dt} = -k(T - A) \quad \text{where } t \text{ is the time in minutes and } k \text{ is a positive constant.}$$

(i) Use differential equations to show that $T = A + Be^{-kt}$ is a solution to the above equation.

1

	1 – demonstrate through integration Many students did not understand to integrate first, rather than start with the solution.
--	---

- (ii) A cup of tea with initial temperature of 90°C is placed in a room in which the surrounding temperature is maintained at 25°C . After 25 minutes, the temperature of the cup of tea is 45°C . How long will it take for the it's temperature to reduce to 30°C ? Answer correct to the nearest minute.

3

$\left. \begin{array}{l} t=0 \\ T=90^{\circ} \\ A=25^{\circ} \end{array} \right\} \begin{array}{l} 90 = 25 + B e^0 \Rightarrow B = 65 \\ \therefore T = 25 + 65 e^{-kt} \end{array}$ $\left. \begin{array}{l} t=25 \\ T=45 \end{array} \right\} \begin{array}{l} 45 = 25 + 65 e^{-25k} \\ e^{-25k} = \frac{4}{13} \Rightarrow k = -\frac{1}{25} \ln \frac{4}{13} \end{array}$ $\therefore T = 25 + 65 e^{\frac{t}{25} \ln \frac{4}{13}}$ $T=30 \Rightarrow 30 = 25 + 65 e^{\frac{t}{25} \ln \frac{4}{13}}$ $e^{\frac{t}{25} \ln \frac{4}{13}} = \frac{5}{65}$ $\frac{t}{25} \ln \frac{4}{13} = \ln \frac{1}{13}$ $t = 25 \frac{\ln \frac{1}{13}}{\ln \frac{4}{13}}$ $\approx 54 \text{ min.}$	1 – finding B 1 – finding k 1 – for final answer Marker's comments: Answered well.
---	---

- (d) (i) Ten friends are going to be divided into groups of 5, 3 and 2 members as part of a competition. In how many ways can the groups be formed?

2

$\binom{10}{5} \binom{5}{3} = 2520$	1 – for one of the combinations 1 – for final answer Marker's comments: Answered well.
-------------------------------------	---

(ii) If n friends are divided into groups of made up of c , k , and r members

where $c + r + k = n$ and $c > k > r$. Explain why

$$\binom{n}{n-k-r} \binom{k+r}{r} = \binom{n}{r} \binom{n-r}{k}.$$

2

A group of n friends are being divided into 3 groups of numbers c, k and r. <u>considering LHS</u>, if we choose the 1st 'c' friends then $k+r$ are left (note: $c = n - k - r$). Then from $k+r$ we choose k, leaving 'r' friends. <u>now considering RHS</u>, we can at first select 'r' friends, leaving '$n-r$' friends. Then from '$n-r$' we choose k friends leaving $c = n - k - r$ friends, resulting in the same number of groupings.	1 – for identifying the selection process on the LHS 1 – for similarly working on the RHS Marker's comments: Well done by those who attempted it
--	---

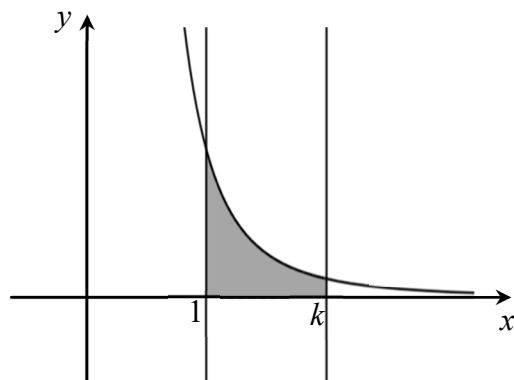
End of Question 3.

Question 4 (13 marks) - Start your work in Question 4 Answer Booklet

- (a) A spherical ball is expanding so that its volume is increasing at the constant rate of 10 mm^3 per seconds. What is the rate of increase of the radius when the surface area is 400 mm^2 ? **2**

$\frac{dv}{dt} = 10 \text{ mm}^3/\text{sec} ; \frac{dr}{dt} = ? \quad SA = 400 \text{ mm}^2$ $V = \frac{4}{3} \pi r^3 \quad SA = 4 \pi r^2$ $\frac{dv}{dr} = 4 \pi r^2 \quad = 400$ $\frac{dr}{dt} = \frac{dr}{dv} \cdot \frac{dv}{dt}$ $= \frac{1}{4 \pi r^2} \cdot 10$ $\therefore \frac{dr}{dt} = \frac{10}{400}$ $= \frac{1}{40} \text{ mm/sec}$	1 – for writing the expression for dr/dt 1 – for final answer Marker's comments: Well done
--	---

- (b) The graph of $y = \frac{1}{x^3} \{x > 0\}$ is shown below. The shaded area is rotated about the y-axis.



- (i) Show that the generated volume in terms of k is $V = \left(2\pi - \frac{2\pi}{k}\right) \text{units}^3$.

4

$y = \frac{1}{x^3} \Rightarrow x^3 = \frac{1}{y}$ $x^2 = y^{-\frac{2}{3}}$ when $x=1 \Rightarrow y=1$ $x=k \Rightarrow y = \frac{1}{k^3}$ $V = \pi \int_a^b x^2 dy + \pi r_1^2 h - \pi r_2^2 h$ $= \pi \int_{\frac{1}{k^3}}^1 y^{-\frac{2}{3}} dy + \pi (1)^2 \left(\frac{1}{k^3}\right) - \pi (1)^2 (1)$ $= \pi \left[3 y^{\frac{1}{3}} \right]_{\frac{1}{k^3}}^1 + \frac{\pi}{k} - \pi$ $= 3\pi \left[1 - \left(\frac{1}{k^3}\right)^{\frac{1}{3}} \right] + \frac{\pi}{k} - \pi$ $= 3\pi - \frac{3\pi}{k} + \frac{\pi}{k} - \pi$ $= \left(2\pi - \frac{2\pi}{k}\right) \text{units}^3$	1 – Integral for the curve with correct limits 1 – Volume for both cylinders 1 – correct integration 1 – for correct simplification Marker's comments: Poorly done . Most of the students couldn't figure out the correct area that is being rotated around y axis .
---	--

- (ii) Explain what happens to the volume as $k \rightarrow \infty$.

1

$\text{As } k \rightarrow \infty, \frac{2\pi}{k} \rightarrow 0 \Rightarrow V = 2\pi \text{units}^3$	1 – for the answer Marker's comments: well done
---	---

- (iii) If the volume of the solid form is $\frac{3\pi}{2} \text{units}^3$, find the value of k .

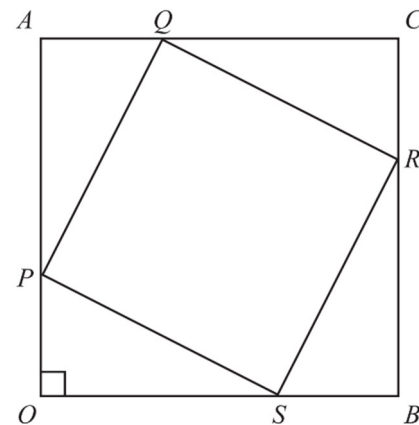
1

$2\pi - \frac{2\pi}{k} = \frac{3\pi}{2}$ $\frac{2\pi}{k} = \frac{\pi}{2}$ $k = 4$ $\therefore k = 6$	1 – for correct answer Marker's comments: very well done
--	--

(c) Consider the square $OACB$ where point O is the origin. Let the position vector of points A and B be defined as $\underline{a}$ and $\underline{b}$ respectively i.e. $\overrightarrow{OA} = \underline{a}$ and $\overrightarrow{OB} = \underline{b}$.

Let points P , Q , R and S be defined so that $\overrightarrow{OP} = k\underline{a}$, $\overrightarrow{AQ} = k\underline{b}$, $\overrightarrow{RC} = k\underline{a}$ and $\overrightarrow{SB} = k\underline{b}$ where $0 \leq k \leq 1$. This means points P , Q , R and S are positioned along their respective sides in equal proportions.

Use vector methods to prove that the size of $\angle PQR = 90^\circ$. **5**



$\begin{aligned}\overrightarrow{PA} &= (1-k)\underline{a} & ; & \quad \overrightarrow{AQ} = k\underline{b} \\ \overrightarrow{QC} &= (1-k)\underline{b} & ; & \quad \overrightarrow{CR} = -k\underline{a} \\ \overrightarrow{PQ} &= \overrightarrow{PA} + \overrightarrow{AQ} & ; & \quad \overrightarrow{QR} = \overrightarrow{QC} + \overrightarrow{CR} \\ &= (1-k)\underline{a} + k\underline{b} & & \quad = (1-k)\underline{b} - k\underline{a} \\ \overrightarrow{PQ} \cdot \overrightarrow{QR} &= [(1-k)\underline{a} + k\underline{b}] \cdot [(1-k)\underline{b} - k\underline{a}] \\ &= (1-k)^2 \underline{a} \cdot \underline{b} - k^2 \underline{a} \cdot \underline{b} \\ \text{Now } \underline{a} \cdot \underline{b} &= 0 & (\text{OACB is a square}) \\ \therefore \overrightarrow{PQ} \cdot \overrightarrow{QR} &= 0 \\ \therefore \angle PQR &= 90^\circ\end{aligned}$	1 – Expression for $\overrightarrow{PQ}$ in terms of $\underline{a}$ and $\underline{b}$ 1 – Expression for $\overrightarrow{QR}$ in terms of $\underline{a}$ and $\underline{b}$ 1 – Using the dot product 1 – Simplifying the dot product 1 – $\underline{a} \cdot \underline{b} = 0$ Marker's comments: well done . Some students struggled in using the properties of square to the dot product of vector PQ and vector QR .
---	---

End of Examination.